Fiscal Year:	FY 2015	**Task Last Updated:**	FY 04/28/2015
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	ISS Microbial Observatory - a Genetic Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Organization Name:	NASA Jet Propulsion Laboratory		
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PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	ISS Microbial Observatory NOTE: Extended to 12/31/2016 per F. Hernandez/ARC (Ed., 4/11/16) NOTE: Extended to 4/30/2016 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fox, George (University of Houston) Pierson, Duane (NASA Johnson Space Center)		
Grant/Contract No.:	19-12829-26 under Task Order NNN13D111T		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As recommended by the National Research Council Decadal Survey, the Jet Propulsion Laboratory (JPL) proposes to establish an International Space Station-Microbial Observatory (ISS-MO) to generate a microbial census of the space station"s surfaces and atmosphere using advanced molecular microbial community analysis techniques, supported by traditional culture-based methods and modern bioinformatic computational modeling. The proposed ISS-MO establishment will lead to long-term, multigenerational studies of microbial population dynamics, addressing recommendation P1 of this solicitation. It will also provide answers to NASA Space Biology Guiding Questions CMM-1, CMM-4, and CMM-5. The ISS-MO project's methodology will serve as the foundation for an extensive microbial census, offering significant insight into spaceflight-induced changes in the populations of beneficial and potentially harmful microbes. Additionally, it will provide NASA with both a mechanistic understanding of these changes (e.g., cataloging population changes and mapping/linking these to environmental niche and genomic changes), as well as insight into practical countermeasures for mitigating risks to humans and environmental systems. The ISS-MO team will use existing ISS sample collection technologies to generate an initial microbial census. Following their return to Earth, samples from the various ISS modules will be analyzed using standardized technologies from the Mars Program-funded projects. The proposed ISS-MO project includes the delivery of a database that compiles the genomic sequences and genetic information for all the microbes encountered within the ISS habitat. Using this data, NASA can more accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to providing microbial profiles, the ISS-MO team will identify which microbial taxa pose particular threats to crew health. Furthermore, specific ISS-MO project aims will enable us to resolve applicable NASA-Human Research Program integrated research plan risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NASA crewmembers are exposed constantly to the dust in a closed system, and have also reported allergies. Subsequent microbiological analyses using traditional methods noticed no anomaly and were also not able to predict the causative microbial agents. The next-generation sequencing study provided NASA with the ability to assess the spectrum of microorganisms associated with the ISS surfaces and would help to predict possible microbial agents that might affect crew health. The results of the present study also exhibited presence of representative human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals and opportunistic fungal pathogens. Furthermore, based on the phylogenetic affiliation it is possible to determine whether these viable microorganisms are hardy and robust that have been previously reported to survive in oligotrophic environments (cleanrooms, hospital, commercial airliner, etc.). The study has improved our understanding of background microbial contamination, thus facilitating the development of biosensors to monitor closed habitats like ISS and future manned missions.
Task Progress:	The safety of the International Space Station (ISS) crewmembers and the maintenance of hardware are the primary goals for monitoring microorganisms in this closed habitat. Together with traditional microbiological methods, state-of-the-art molecular technologies were utilized to catalogue the microbiota associated with the ISS environment as well as samples from multiple Earth cleanroom environments. Surface and air samples were collected and analyzed. In this study, traditional cultivation, adenosine triphosphate (ATP) assay, and propidium monoazide-quantitative polymerase chain reaction (PMA-qPCR) method were employed to quantify the microbial populations. In addition, the next-generation sequencing analyses were used to elucidate the microbial diversity and clarify difference between the microbiomes of ISS and Earth cleanrooms. The statistical analysis of these microbiomes showed that three bacterial phyla dominated both in ISS and Earth cleanrooms, but varied in their abundances. While members of Actinobacteria were predominant on ISS, Proteobacteria, the least abundant on ISS, dominated the Earth cleanrooms. Alpha diversity estimators indicated a significant drop in diversity associated with PMA treated samples. On average, PMA treated samples had 48% fewer operational taxonomic unit (OTU) than their respective untreated samples (paired T-test P= 0.004). This consistent reduction in microbial diversity of PMA samples was also true for Faith’s diversity, Chao1, and Shannon diversity measures, and was also constant with the depletion of many taxa identified by differential abundance analysis. To better characterize the shared community composition among samples, beta-diversity metrics analysis were conducted. A significant difference was noticed between ISS and cleanroom samples, i.e., samples from the same site are highly similar microbiome composition regardless of PMA treatment. At the OTU level, the microbial community composition is strongly associated with sampling site, and the reduced overall diversity of PMA treatment does not dramatically impact the composition relative to sample origin. Results of the study indicate significant differences between ISS and Earth cleanroom microbiomes in terms of community structure and composition.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Checinska A, Probst AJ, White JR, Vaishampayan P, Perry J, Venkateswaran K. "Is the International Space Station Microbiome Different from the Earth Cleanrooms?" Presentation at the 115th General Meeting of the American Society for Microbiology, New Orleans, LA, May 29-June 2, 2015. 115th General Meeting of the American Society for Microbiology, New Orleans, LA, May 29-June 2, 2015. , May-2015
Articles in Peer-reviewed Journals	Venkateswaran K, La Duc MT, Horneck G. "Microbial existence in controlled habitats and their resistance to space conditions." Microbes Environ. 2014 Sep 17;29(3):243-9. http://dx.doi.org/10.1264/jsme2.ME14032 ; PubMed PMID: 25130881; PubMed Central PMCID: PMC4159035 , Sep-2014